Signals and Systems Tutorial-02 Problems

Problems On Periodicity, Delta Function Properties, Even and Odd, Power and Energy of Signals

1) Solve the Following Periodicity Problem as Shown Below:

A signal $g(t)$ that is a linear combination of two periodic signals, $x_1(t)$ with fundamental period T_1 and $x_2(t)$ with fundamental period T_2 as $g(t) = ax_1(t) + bx_2(t)$ is periodic iff

$$\frac{T_1}{T_2} = \frac{m}{n} = a\ rational\ number$$

Prove that the fundamental period of $g(t)$ is given by $nT_1 = mT_2$ provided that the values of m and n are chosen such that the greatest common divisor (gcd) between m and n is 1.

2) Find the Periodicity of the Following Problems as Shown Below:

Determine if the following continuous time (CT) signals are periodic. If true, calculate their fundamental period T_0. If the signals are aperiodic, provide proper justification for the same.

(a) $sin\ (-\frac{5\pi}{8}t + \frac{\pi}{2})$

(b) $|sin\ (-\frac{5\pi}{8}t + \frac{\pi}{2})|$

(c) $sin(\frac{6\pi t}{7}) + 2\ cos(\frac{3t}{5})$

(d) $e^{j(5t+\frac{\pi}{4})}$

(e) $e^{j\frac{3\pi}{8}t} + e^{\frac{\pi t}{86}}$

(f) $e^{j\frac{3\pi}{8}t} + e^{\frac{\pi}{86}}$

(g) $2cos(\frac{4\pi t}{5})\ sin^2(\frac{16t}{3})$

(h) $1 + sin(20t) + cos(30t + \frac{\pi}{3})$

3) Solve the Following Problems Using Delta Function Properties

(a) $\frac{5-jt}{7+t^2}\delta(t)$

(b) $\int_{-\infty}^{\infty}(t+5)\delta(t-2)\,dt$

(c) $\int_{-\infty}^{\infty} e^{j(0.5\pi\omega+2)}\delta(\omega-5)\,d\omega$

(d) $\int_{-\infty}^{\infty} e^{-2t}\delta(-2t+1)\,dt$

(e) $\int_{-\infty}^{\infty} e^{-t}\delta(2t-2)\,dt$

(f) $\int_{0}^{\infty} e^{-\alpha t^2}\delta(t+5)\,dt$

(g) $\int_{-2}^{11} u(4-t)\,dt$

(h) $\int_{-\infty}^{\infty} e^{-18t}u(t)\delta(10t-2)\,dt$

4) Solve the Following periodicity Problem as Shown Below:

Determine if the following signals are periodic. If they are periodic, find the fundamental period:

(a) $x[n] = (-1)^{n^2}$

(b) $x[n] = (-1)^n$

(c) $x[n] = cos(2\pi n)$

(d) $x[n] = cos(2n)$

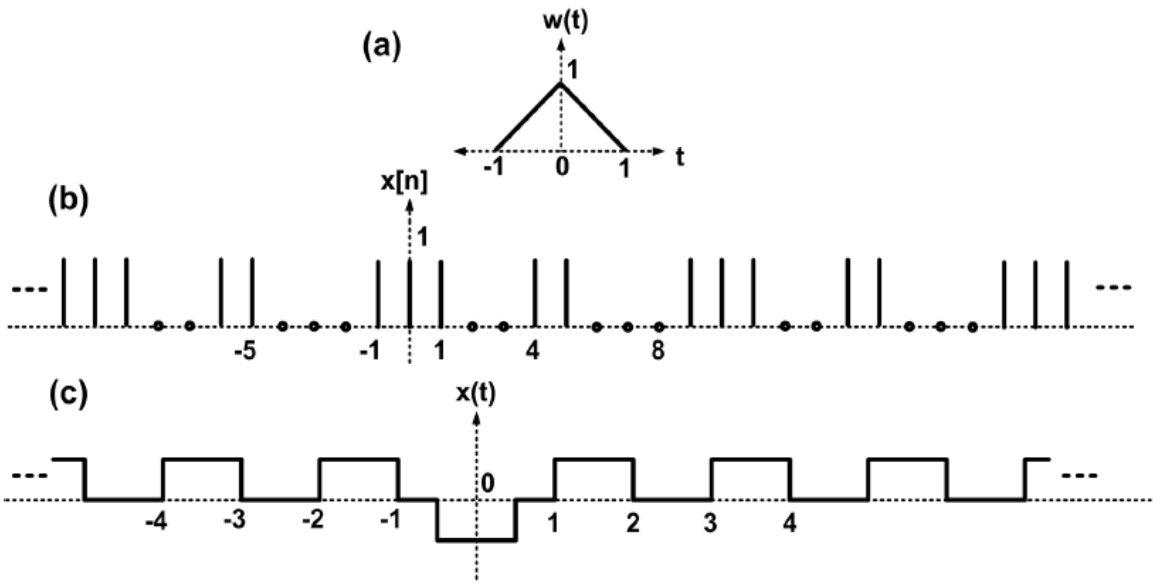
(e) $x(t) = (cos(2\pi t))^2$

(f) $x(t) = \sum_{k=-5}^{5} w(t-2k)$ for $w(t)$ depicted in Fig. 1(a)

(g) $x(t) = \sum_{k=-\infty}^{\infty} w(t-3k)$ for $w(t)$ depicted in Fig. 1(a)

(h) $x[n]$ depicted in Fig. 1(b)

(i) $x[n]$ depicted in Fig. 1(c)

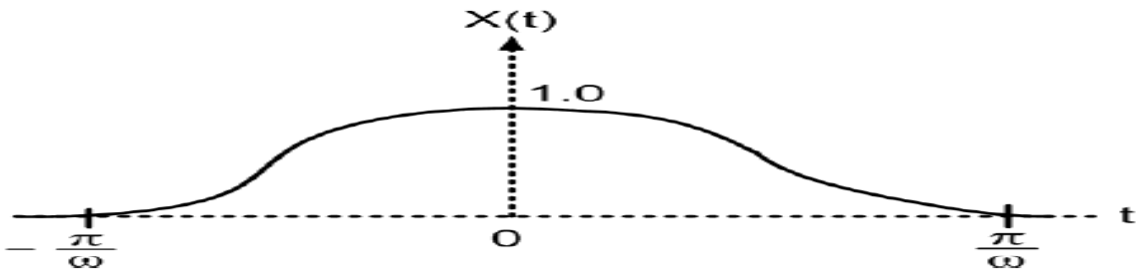


5) Solve the Given Energy Problem as Shown Below:

The raised cosine pulse $x(t)$ shown in Fig. 2 is defined as:

$$x(t) = \begin{cases} \frac{1}{2}[cos(\omega t) + 1], & \text{if } -\frac{\pi}{\omega} \leq t \leq \frac{\pi}{\omega} \\ 0, & \text{otherwise} \end{cases}$$

Determine the total energy of $x(t)$.



6) Solve the Given Problem as Shown Below:

An arbitrary real-valued continuous time signal may be expressed as $x(t) = x_e(t) + x_o(t)$, where $x_e(t)$ and $x_o(t)$ are the even and odd components of $x(t)$ respectively. The signal $x(t)$ occupies the entire interval $-\infty < t < \infty$. Show that the energy of the signal $x(t)$ is equal to the sum of the energy of even component $x_e(t)$ and the energy of odd component $x_o(t)$.

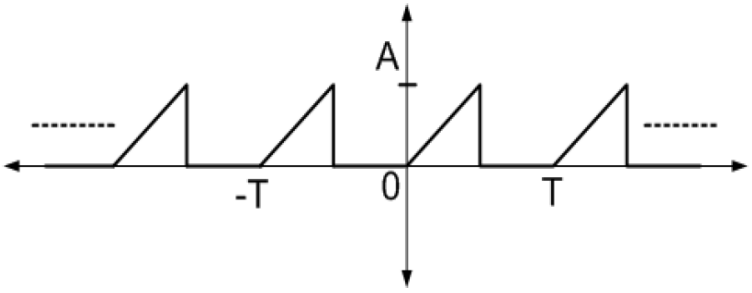
7) Solve the Given Problem as Shown Below:

Find the strength of the following impulses:

(a) $-3\delta(-4t)$

(b) $5\delta(3(t-1))$

8) Shows a periodic 50% duty cycle saw tooth wave $x(t)$ as shown below with peak amplitude A. Determine the energy and power of $x(t)$



9) Solve the Given Problem as Shown Below:

Let $x[n]$ be a discrete-time signal, and let

$$y_1[n] = x[2n] \text{ and } y_2[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}.$$

The signals $y_1[n]$ and $y_2[n]$ respectively represent in some sense the speeded up and slowed down versions of $x[n]$. However, it should be noted that the discrete-time notions of speeded up and slowed down have subtle differences with respect to their continuous-time counterparts. Consider the following statements:

(1) If $x[n]$ is periodic, then $y_1[n]$ is periodic.
(2) If $y_1[n]$ is periodic, then $x[n]$ is periodic.
(3) If $x[n]$ is periodic, then $y_2[n]$ is periodic.
(4) If $y_2[n]$ is periodic, then $x[n]$ is periodic.

For each of these statements, determine whether it is true, and if so, determine the relationship between the fundamental periods of the two signals considered in the statement. If the statement is not true, produce a counterexample to it.

Bonus Question

Show that if $x(t)$ is periodic with period T, then

$$\left| \int_0^T x(t)\, dt \right| \leq \sqrt{PT}$$

where P is the average power of the signal.